Calculus II Lesson 10

1) $\sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^n = \frac{1}{6} + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \cdots$

Note: $\sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^n = \sum_{n=1}^{\infty} ar^n$ is a geometric series;

where a is the first term and r is the common ratio.

a = first term = 1/6

r = common ratio = 1/6

$$\text{Let } S = \frac{1}{6} + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \left(\frac{1}{6}\right)^5 + \ldots$$

$$r \cdot S = \frac{1}{6} \cdot S = \frac{1}{6} \cdot \left[\frac{1}{6} + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \left(\frac{1}{6}\right)^5 + \ldots\right] = \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \left(\frac{1}{6}\right)^5 + \ldots$$

$$S - \frac{1}{6} \cdot S = \left[\frac{1}{6} + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \left(\frac{1}{6}\right)^5 + \ldots\right] - \left[\left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^3 + \left(\frac{1}{6}\right)^4 + \left(\frac{1}{6}\right)^5 + \ldots\right]$$

a) $\frac{5}{6}S =$ ______?______

b) $\sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^n = S =$ ______?______

c) Use the Geometric Series Test to explain why series $\sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^n$ converges or diverges.

Geometric Series Test for Infinite Series $\sum_{n=1}^{\infty} ar^n$

a) Geometric Series $\sum_{n=1}^{\infty} ar^n$ converges to $\frac{a}{1-r}$ if r is between -1 and 1.

b) Geometric Series $\sum_{n=1}^{\infty} ar^n$ diverges if r is not between -1 and 1.

2) $\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n=\left(\frac{1}{2}\right)^0+\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^5+\ldots$

Note: $\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n$ is a Geometric Series with $a=$ first term $=\left(\frac{1}{2}\right)^0=1$ and $r=$ common ratio $=\frac{1}{2}$

Let $S=\left(\frac{1}{2}\right)^0+\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^5+\ldots$

$$r\cdot S=\frac{1}{2}\cdot S=\frac{1}{2}\cdot\left[\left(\frac{1}{2}\right)^0+\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^5+\ldots\right]$$

$$\frac{1}{2}\cdot S=\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\ldots$$

$$S-\frac{1}{2}\cdot S=\left[\left(\frac{1}{2}\right)^0+\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^5+\ldots\right]-\left[\left(\frac{1}{2}\right)^1+\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\ldots\right]$$

a) $S = \underline{\quad ? \qquad\qquad}$

b) Use the Geometric Series Test to explain why series $\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n$ converges or diverges.

Geometric Series Test for Infinite Series $\sum_{n=1}^{\infty} ar^n$

a) Geometric Series $\sum_{n=1}^{\infty} ar^n$ converges to $\frac{a}{1-r}$ if r is between -1 and 1.

b) Geometric Series $\sum_{n=1}^{\infty} ar^n$ diverges if r is not between -1 and 1.

3) $\sum_{n=1}^{\infty} 4 \cdot \left(\frac{1}{3}\right)^n = 4 \cdot \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$

$$\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 + \left(\frac{1}{3}\right)^5 + \ldots$$

Note: $\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$ is a Geometric Series with a = first term = $\left(\frac{1}{3}\right)^1 = \frac{1}{3}$ and r = common ratio = $\frac{1}{3}$

Let $S = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 + \left(\frac{1}{3}\right)^5 + \ldots$

$$r \cdot S = \frac{1}{3} \cdot S = \frac{1}{3} \cdot \left[\left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 + \left(\frac{1}{3}\right)^5 + \ldots\right]$$

$$\frac{1}{3} \cdot S = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^5 + \ldots$$

$$S - \frac{1}{3} \cdot S = \left[\left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 + \left(\frac{1}{3}\right)^5 + \ldots\right] - \left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^5 + \ldots\right]$$

a) $\frac{2}{3}S =$ ______?______ $\Rightarrow$ $S =$ ______?______

b) $4 \cdot \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n =$ ______?______

c) Use the Geometric Series Test to explain why series $4 \cdot \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$ converges or diverges.

Geometric Series Test for Infinite Series $\sum_{n=1}^{\infty} ar^n$

a) Geometric Series $\sum_{n=1}^{\infty} ar^n$ converges to $\frac{a}{1-r}$ if r is between -1 and 1.

b) Geometric Series $\sum_{n=1}^{\infty} ar^n$ diverges if r is not between -1 and 1.

4) Find the sum $\sum_{n=1}^{\infty}(1.2)^n$ = ____?____

Use the Geometric Series Test to explain why the series $\sum_{n=1}^{\infty}(1.2)^n$ converges or diverges.

Geometric Series Test for Infinite Series $\sum_{n=1}^{\infty} ar^n$

a) Geometric Series $\sum_{n=1}^{\infty} ar^n$ converges to $\dfrac{a}{1-r}$ if r is between -1 and 1.

b) Geometric Series $\sum_{n=1}^{\infty} ar^n$ diverges if r is not between -1 and 1.

5) Find $\sum_{n=0}^{\infty} 3 \cdot (0.7)^n = 3 \cdot \sum_{n=0}^{\infty} (0.7)^n =$ ______ ? ____________

Note: $\sum_{n=0}^{\infty} (0.7)^n$ is a Geometric Series with $a =$ first term $= (0.7)^0 = 1$ and $r =$ common ratio $= 0.7$

a) $\sum_{n=0}^{\infty} (0.7)^n =$ ______ ? ____________

b) $3 \cdot \sum_{n=0}^{\infty} (0.7)^n =$ ______ ? ____________

c) Use the Geometric Series Test to explain why the series $3 \cdot \sum_{n=0}^{\infty} (0.7)^n$ converges or diverges.

6) Find all values of x for which the series $\sum_{n=1}^{\infty}(2x)^n$ converges.

7) Find all values of x for which the series $\sum_{n=1}^{\infty}(3x-1)^n$ converges.

8) Find all values of x for which the series $\sum_{n=1}^{\infty}\left(\frac{3}{4}x\right)^n$ converges.